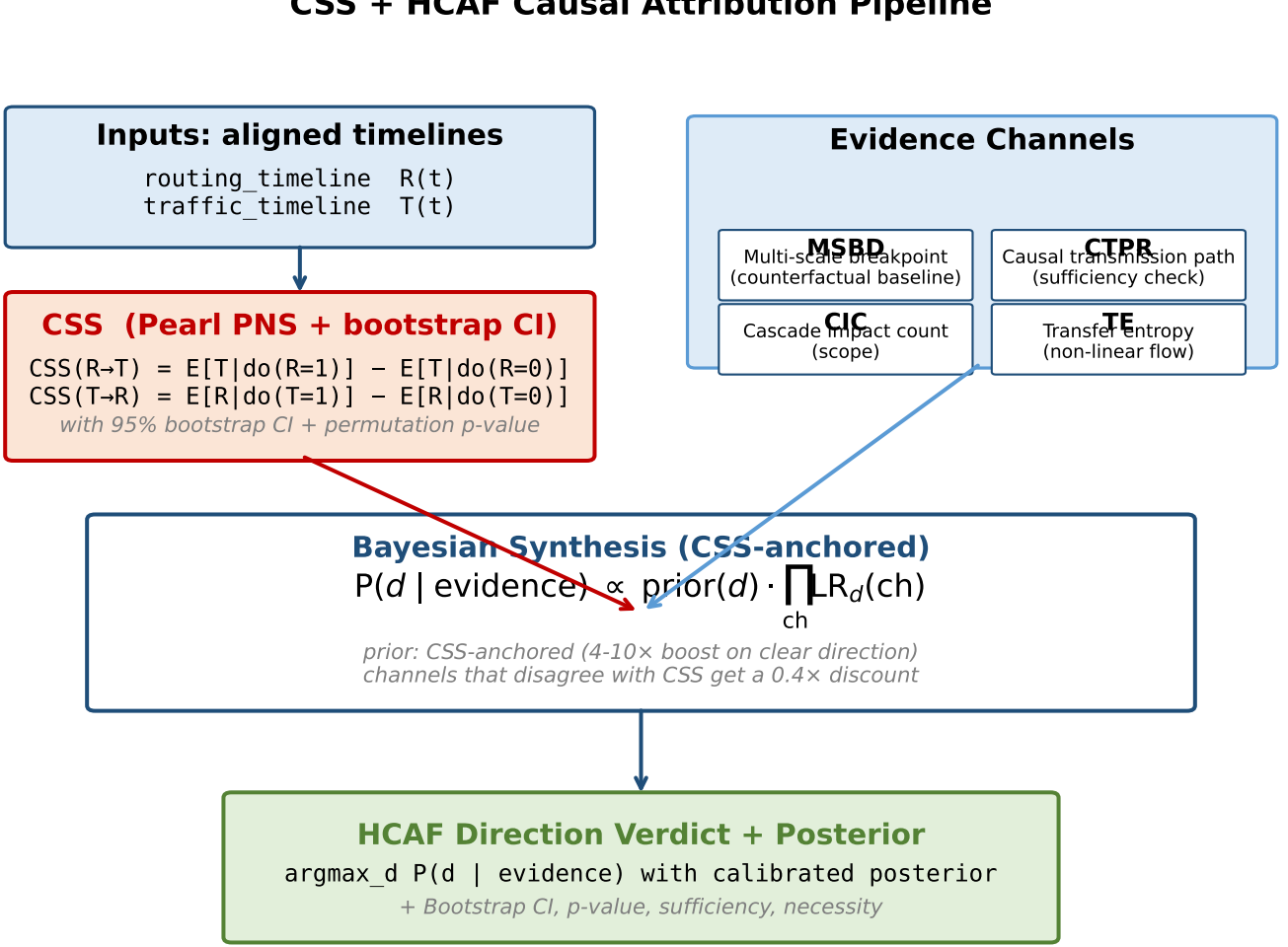
CSS + HCAF Causal Attribution Pipeline
Inputs: aligned timelines
routing_timeline R(t)
traffic_timeline T(t)
Evidence Channels
MSBD
Multi-scale breakpoint (counterfactual baseline)
CTPR
Causal transmission path (sufficiency check)
CIC
Cascade impact count (scope)
TE
Transfer entropy (non-linear flow)
CSS (Pearl PNS + bootstrap CI)
CSS(R→T) = E[T|do(R=1)] − E[T|do(R=0)]
CSS(T→R) = E[R|do(T=1)] − E[R|do(T=0)]
with 95% bootstrap CI + permutation p-value
Bayesian Synthesis (CSS-anchored)
P(d | evidence) ∝ prior(d) · ∏ LR_d(ch)
ch
prior: CSS-anchored (4-10× boost on clear direction)
channels that disagree with CSS get a 0.4× discount
HCAF Direction Verdict + Posterior
argmax_d P(d | evidence) with calibrated posterior
+ Bootstrap CI, p-value, sufficiency, necessity